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METHOD FOR SEMICONDUCTOR MANUFACTURING

Abstract

A method of processing a substrate includes forming a recess by etching a sacrificial layer, forming an inner spacer in the recess, and forming a source/drain region. The sacrificial layer is between a lower nanosheet and an upper nanosheet. The source/drain region is formed adjacent the lower nanosheet, the inner spacer, and the upper nanosheet. The method further includes removing the sacrificial layer with an etch process and performing an oxidation on a portion of the source/drain region through the inner spacer.

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Background/Summary

TECHNICAL FIELD

[0001] The present invention relates generally to semiconductor manufacturing, and, in particular embodiments, to methods of processing a substrate.

BACKGROUND

[0002] Generally, a semiconductor device, such as an integrated circuit (IC) is fabricated by sequentially depositing and patterning layers of dielectric, conductive, and semiconductor materials over a substrate to form a network of electronic components and interconnect elements (e.g., transistors, resistors, capacitors, metal lines, contacts, and vias) integrated in a monolithic structure. Many of the processing steps used to form the constituent structures of semiconductor devices is performed using various deposition and etch techniques such as plasma processes.

[0003] The semiconductor industry has traditionally followed Moore's Law, which was initially based on the observation that the number of transistors on a chip doubles approximately every two years, leading to a cadence of shrinking feature sizes (also referred to as "scaling") along with improvements in performance and reductions in costs. However, as transistor features approached atomic dimensions, maintaining this pace has become increasingly challenging. As a result, the scaling cadence has evolved from a strict focus on gate length reduction to a more complex one incorporating innovations in 3D structures, new materials, and integration methods.

[0004] Nanosheet transistors have emerged as a promising advancement in this endeavor, representing a significant evolution from their planar and FinFET predecessors. Nanosheet transistors, characterized by their multiple horizontal sheets, or 'nanosheets', of channel material stacked vertically, allow for enhanced control over the current flow and offer the potential for further scaling beyond the limits of conventional FinFET architectures. These devices can effectively maintain excellent electrostatic control over a channel and enable gate-all-around (GAA) configurations, which are imperative for next-generation integrated circuits and high-performance computing applications. Despite the advantages of nanosheet transistors, challenges remain in fabrication techniques, uniformity across large wafers, thermal management, and integration with existing manufacturing processes.

SUMMARY

[0005] In accordance with an embodiment, a method of processing a substrate includes: forming a recess by etching a sacrificial layer, the sacrificial layer being between a lower nanosheet and an upper nanosheet; forming an inner spacer in the recess; forming a source/drain region adjacent the lower nanosheet, the inner spacer, and the upper nanosheet; removing the sacrificial layer with an etch process; and performing an oxidation on a portion of the source/drain region through the inner spacer.

[0006] In accordance with another embodiment, a method of processing a substrate includes: forming a first layer stack and a second layer stack, the first layer stack including alternating layers of first nanosheets and first sacrificial layers, the second layer stack including alternating layers of second nanosheets and second sacrificial layers, the second layer stack being wider than the first layer stack; performing a first channel release process on the first layer stack and the second layer stack, the first channel release process removing the first sacrificial layers from the first layer stack and a portion of the second sacrificial layers from the second layer stack; performing an oxidation on a portion of a source/drain region, the source/drain region being adjacent the first layer stack; and performing a second channel release process on the second layer stack, the second channel release process removing the remaining portion of the second sacrificial layers.

[0007] In accordance with yet another embodiment, a method of processing a substrate includes: forming a first layer stack and a second layer stack, the first layer stack and the second layer stack each including alternating layers of silicon (Si) layers and silicon-germanium (SiGe) layers, the first layer stack being wider than the second layer stack; forming indents between the Si layers of the first layer stack and the second layer stack by etching a portion of the SiGe layers selectively to

the Si layers; forming respective inner spacers in the indents of the first layer stack and the second layer stack; epitaxially growing a first source/drain region adjacent the first layer stack and a second source/drain region adjacent the second layer stack; removing a portion of the SiGe layers of the first layer stack and the SiGe layers of the second layer stack with a first etch process; forming oxidized regions in the second source/drain region through respective inner spacers of the second layer stack; and removing the remaining portion of the SiGe layers of the first layer stack with a second etch process.

[0008] It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the disclosure, as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] For a more complete understanding of the present invention, and the advantages thereof, reference is now made to the following descriptions taken in conjunction with the accompanying drawings, in which:

[0010] FIGS. 1A and 1B illustrate cross-sectional views of an example substrate;

[0011] FIGS. 2, 3A, 3B, 4A, 4B, 5, 6, 7A, 7B, 8, 9A, 9B, 10A, 10B, 11A, and 11B illustrate cross-sectional views of intermediate stages of a manufacturing process for a semiconductor structure, in accordance with some embodiments;

[0012] FIG. 11C illustrates a graph of etched amounts of oxidized silicon-germanium (SiGe) versus germanium percentage of the SiGe, in accordance with some embodiments;

[0013] FIG. 11D illustrates experimental results for decrease in etching of oxidized silicon-germanium (SiGe) versus germanium percentage of the SiGe, in accordance with some embodiments;

[0014] FIGS. 12A, 12B, 13A, and 13B illustrates cross-sectional views of intermediate stages of a manufacturing process for a semiconductor structure, in accordance with some embodiments;

[0015] FIG. 14 illustrates a flow chart diagram of a method of processing a substrate, in accordance with some embodiments;

[0016] FIG. 15 illustrates a flow chart diagram of a method of processing a substrate, in accordance with some embodiments; and

[0017] FIG. 16 illustrates a flow chart diagram of a method of processing a substrate, in accordance with some embodiments.

[0018] Corresponding numerals and symbols in the different figures generally refer to corresponding parts unless otherwise indicated. The figures are drawn to clearly illustrate the relevant aspects of the embodiments and are not necessarily drawn to scale. The edges of features drawn in the figures do not necessarily indicate the termination of the extent of the feature.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

[0019] The making and using of various embodiments are discussed in detail below. It should be appreciated, however, that the various embodiments described herein are applicable in a wide variety of specific contexts. The specific embodiments discussed are merely illustrative of specific ways to make and use various embodiments, and should not be construed in a limited scope.

[0020] Methods of reducing or preventing damage to source/drain regions (in other words, to source regions and drain regions) during channel release processes. Generally, fabricating complicated structures for advanced semiconductor devices, for example 3-D devices such as gate-all-around field effect transistors (GAAFETs, also referred to as nanosheet or nanowire FETs) and stacked FETs (also referred to as complementary FETs or CFETs), may require laterally removing materials to selectively expose a portion of the underlying structure. However, it may be challenging to achieve sufficient etch selectivity in certain processes due to the small scale and

design complexity of the features to be fabricated.

[0021] One example of such difficult etch processes is a channel release process for a nanosheet/nanowire p-channel FET (pFET), where the insufficient etch selectivity may cause significant damage to source/drain regions. Commonly, as illustrated in FIG. 1A, a layer stack of silicon (Si) nanosheets **10** and sacrificial layers **20** is formed to provide multiple Si nanosheet/nanowire channels. Source/drain materials **30** are then epitaxially grown, such as from the tips of the Si nanosheet/nanowire and/or from an adjacent recess in an Si substrate. In typical examples, the source/drain materials **30** comprise boron-doped silicon-germanium (B-doped SiGe) and the sacrificial layers **20** comprise un-doped SiGe.

[0022] During the channel release process, the sacrificial layers **20** are removed by dry etching. An inner spacer **40** is used to separate the sacrificial layers **20** from the source/drain materials **30** in order to protect the source/drain materials **30** during the channel release process. However, as illustrated in FIG. 1B with arrows, the etch gas used in the conventional channel release process may penetrate the inner spacer **40** to reach and damage the source/drain materials **30**. Although the inner spacer **40** may be in principle chemically resistant to the etch gas and able to provide etch selectivity, the penetration of the etch gas may still occur because the inner spacer **40** is typically scaled to a small thickness and made porous to reduce its dielectric constant in order to minimize the parasitic capacitance caused by the inner spacer **40**. Therefore, a new solution for channel release processes that reduce or prevent source/drain region damage is desirable.

[0023] According to one or more embodiments of the present disclosure, methods of semiconductor manufacturing include selective oxidation of portions of source/drain regions and subsequent channel release selective to the oxidized portions of the source/drain regions. Surface layers of source/drain regions may be selectively oxidized through inner spacers to reduce or prevent damage to the source/drain regions that may occur during over-etching. For example, during channel release processes for devices including both short channel and long channel nanosheet stacks, over-etching may be used in order to remove sacrificial material from the long channel nanosheet stacks after most or all sacrificial material has already been removed from the short channel nanosheet stacks. It may be advantageous to selectively oxidize surface portions of source/drain regions adjacent to the short channel nanosheet stacks and then perform a channel release process with selectivity to the oxidized source/drain material, such as during over-etching for removing remaining sacrificial material from the long channel nanosheet stacks. This selective oxidation and subsequent selective channel release may decrease or prevent damage to source/drain regions without negatively affecting the channel release process performance, thereby increasing device yield and device performance. The selective oxidation and subsequent selective channel release may be advantageous by increasing the margin of over-etching for channel release processes, thereby reducing channel release process marginality and improving device yield. The steps of the methods for selective oxidation and subsequent selective channel release are compatible with existing process flows and may be performed in a same chamber (in other words, in situ). The selective oxidation and channel release processes may be performed without plasma.

[0024] Embodiments of the disclosure are described in the context of the accompanying drawings. An embodiment of a manufacturing process for a semiconductor structure will be described using FIGS. 2, 3A, 3B, 4A, 4B, 5, 6, 7A, 7B, 8, 9A, 9A, 10A, 10B, 11A, 11B, 12A, 12B, 13A, and 13B. An embodiment related to a relationship between etched amounts of oxidized silicon-germanium (SiGe) versus germanium percentage of the SiGe will be described using FIG. 11C. Experimental results for decrease in etching of oxidized silicon-germanium (SiGe) versus germanium percentage of the SiGe will be described using FIG. 11D. Embodiments of methods for processing substrates will be described using FIGS. 14, 15, and 16. Although this disclosure describes embodiments of selective oxidation of source/drain regions subsequent selective channel release in GAAFET applications, the methods of selective source/drain oxidation and selective channel release may also be applied in various other applications such as stacked FET applications.

[0025] FIGS. 2, 3A, 3B, 4A, 4B, 5, 6, 7A, 7B, 8, 9A, 9A, 10A, 10B, 11A, 11B, 12A, 12B, 13A, and 13B illustrate cross-sectional views of a semiconductor device (e.g., a GAAFET device) at intermediate stages of fabrication as an exemplary process flow including a selective oxidation of source/drain regions and subsequent channel release process, in accordance with some embodiments.

[0026] FIG. 2 illustrates a cross-sectional view of a semiconductor structure **100** during fabrication. The semiconductor structure **100** may have undergone a number of steps of processing following, for example, a conventional process. As an example, the semiconductor structure **100** comprises a substrate **110** in which various device regions are formed. At this stage, the substrate **110** may include isolation regions such as shallow trench isolation (STI) regions as well as other regions formed therein.

[0027] In various embodiments, the substrate **110** comprises a semiconductor substrate. In one or more embodiments, the substrate **110** is a silicon wafer or a silicon-on-insulator (SOI) wafer. In certain embodiments, the substrate **110** comprises a germanium wafer, silicon-germanium wafer, silicon carbide wafer, gallium arsenide wafer, gallium nitride wafer and other compound semiconductors. In other embodiments, the substrate **110** comprises heterogeneous layers such as silicon-germanium on silicon, gallium nitride on silicon, silicon carbon on silicon, as well layers of silicon on a silicon or SOI substrate.

[0028] As further illustrated in FIG. 2, a plurality of nanosheets **130** (also referred to as nanosheet layers) are formed over the substrate **110**. Specifically, the nanosheets **130** are embedded in a different material. The nanosheets **130** may be spaced apart from each other by one of a plurality of sacrificial layers **120**. Thus, a layer stack comprising alternating layers of the sacrificial layers **120** and the nanosheets **130** is over the substrate **110**. It should be noted that while three layers of the nanosheets **130** are depicted in FIG. 2, the number of layers is not limited. In various embodiments, at the end of fabrication the nanosheets **130** form channels of a transistor device, while the sacrificial layers **120** will be removed in a later step of fabrication to free up a void space for the formation of a gate dielectric and gate electrode. In various embodiments, the nanosheets **130** have respective thicknesses of a few nanometers to tens of nanometers. For example, in one embodiment the nanosheets **130** have respective thicknesses in a range of 1 nm to 20 nm. In another embodiment, the nanosheets **130** have respective thicknesses in a range of 1 nm to 10 nm. However, the nanosheets **130** may have any suitable thicknesses.

[0029] In various embodiments, the sacrificial layers **120** comprise silicon-germanium (SiGe) and the nanosheets **130** comprise silicon (e.g., crystalline silicon). In other embodiments, the sacrificial layers **120** comprise silicon and the nanosheets **130** comprise silicon-germanium. For example, n-type field effect transistors and p-type field effect transistors may be formed with different types of materials. n-FETs may be fabricated with using nanosheets **130** having high electron mobility while p-FETs may be fabricated with using nanosheets **130** having high hole mobility. In certain embodiments, the nanosheets **130** are selected to be a material selected from Groups III-V of the periodic table and the sacrificial layer **120** is selected to be a material from groups II-VI or group IV of the periodic table. In one or more embodiments, the sacrificial layer **120** comprises silicon-germanium with a ratio of $\text{Si.sub.1-xGe.sub.x}$ where x is less than 0.25.

[0030] In some embodiments, a layer stack of the nanosheets **130** and the sacrificial layer **120** is formed by deposition processes, for example, epitaxially by a chemical vapor deposition (CVD) method. In various embodiments, each layer of the sacrificial layers **120** and the nanosheets **130** may be a few to several nanometers in thickness. In one embodiment, each layer of the sacrificial layers **120** has a thickness in a range of 5 nm and 20 nm and each layer of the nanosheets **130** has a thickness in a range of 1 nm and 10 nm. However, the sacrificial layers **120** and the nanosheets **130** may have any suitable thicknesses.

[0031] As further illustrated in FIG. 2, in some embodiments a dielectric blocking layer **140** is over the alternating layer stack of the nanosheets **130** and the sacrificial layer **120**. In various

embodiments, the dielectric blocking layer **140** is an oxide layer. The dielectric blocking layer **140** may be formed by one or more suitable deposition processes, such as by a CVD method. The dielectric blocking layer **140** may be used as an etch stop layer, such as for a dummy gate removal process, and may be optional.

[0032] FIGS. **3A** and **3B**, following from FIG. **2**, illustrate cross-sectional views of the semiconductor structure **100** after forming dummy gate structures over the layer stack. FIG. **3A** illustrates a short channel region **100A** of the semiconductor structure **100** that may be used for the formation of short channel devices (e.g., short channel GAAFETs or the like), and FIG. **3B** illustrates a long channel region **100B** of the semiconductor structure **100** that may be used for the formation of long channel devices (e.g., long channel GAAFETs or the like).

[0033] In various embodiments, a dummy stack comprising a dummy material is formed over the stack of the nanosheets **130** and the sacrificial layer **120** in the short channel region **100A** and the long channel region **100B**. The dummy stack is patterned to form dummy gates **150**. FIG. **3A** illustrates a feature of two fins for dummy gates **150** and FIG. **3B** illustrates a feature of one fin for a dummy gate **150** as examples. However, any suitable number of dummy gates **150** may be formed for any suitable number of fins. The dummy gates **150** comprise a suitable material such as polysilicon or amorphous silicon. The dummy gates **150** may be deposited using deposition techniques such as vapor deposition including chemical vapor deposition (CVD), physical vapor deposition (PVD), as well as other plasma processes such as plasma enhanced CVD (PECVD), sputtering, the like, or a combination thereof. In some embodiments, the dummy gates **150** have respective thicknesses of 50 nm to 500 nm.

[0034] Still referring to FIGS. **3A** and **3B**, in some embodiments a hard mask layer is formed before the patterning of the dummy gates **150** to form a hard mask **160**. In various embodiments, the hard mask **160** comprises silicon oxide (SiO), silicon dioxide (SiO.sub.2), silicon nitride (SiN), silicon carbonitride (SiCN), silicon oxycarbide (SiOC), silicon oxynitride (SiON), silicon carbide (SiC), amorphous silicon, polycrystalline silicon, titanium nitride (TiN), the like, or a combination thereof. Further, the hard mask **160** may be a stacked hard mask comprising, for example, two or more layers, each of which is formed with a different material. In an example, the first hard mask of the hard mask **160** comprises a metal-based layer such as titanium nitride (TiN), titanium (Ti), tantalum nitride (TaN), tantalum (Ta), tungsten (W) based compounds, ruthenium (Ru) based compounds, or aluminum (Al) based compounds, and the second hard mask of the hard mask **160** comprises a dielectric layer such as silicon oxide (SiO), silicon dioxide (SiO.sub.2), silicon nitride (SiN), silicon carbonitride (SiCN), silicon oxycarbide (SiOC), silicon oxynitride (SiON), silicon carbide (SiC), amorphous silicon, or polycrystalline silicon. The hard mask **160** may be deposited using deposition techniques such as vapor deposition including chemical vapor deposition (CVD), physical vapor deposition (PVD), as well as other plasma processes such as plasma enhanced CVD (PECVD), sputtering, and other processes. In various embodiments, the hard mask **160** has a thickness in a range of 5 nm to 50 nm. In some embodiments, the hard mask **160** is patterned over the layer stack comprising alternating layers of the sacrificial layers **120** and the nanosheets **130**, and the dummy gates **150** are not present.

[0035] After patterning to form the dummy gate **150** and the hard mask **160**, a sidewall spacer layer **168** is deposited over the alternating layer stack of the nanosheets **130** and the sacrificial layer **120**. In various embodiments, the optional dielectric blocking layer **140** may be etched prior to depositing the sidewall spacer layer **168**. In various embodiments, the sidewall spacer layer **168** comprise a dielectric material comprising an oxide or a nitride. In some embodiments, the sidewall spacer layer **168** comprises silicon-containing dielectric materials such as silicon oxide (SiO), silicon dioxide (SiO.sub.2), silicon oxynitride (SiON), silicon oxycarbonitride (SiOCN), silicon boron carbonitride (SiBCN), the like, or a combination thereof. The sidewall spacer layer **168** may be deposited using deposition techniques such as vapor deposition including chemical vapor deposition (CVD), physical vapor deposition (PVD), as well as other plasma processes such as

plasma enhanced CVD (PECVD), atomic layer deposition (ALD), sputtering, and other processes. In various embodiments, the sidewall spacer layer **168** has a thickness in a range of 1 nm to 10 nm. In certain embodiments, the sidewall spacer layer **168** is a stacked layer comprising, for example, two or more layers using two different materials.

[0036] In some embodiments, the dummy gate structures (including dummy gates **150** and sidewall spacer layer **168**) in the short channel region **100A** have respective widths $W_{sub.1}$ in a range of 10 nm to 50 nm, and the dummy gate features in the long channel region **100B** have respective widths $W_{sub.2}$ in a range of 80 nm to 120 nm. These widths correspond to channel widths (e.g., widths of the nanosheets **130**, which may be used to form channels in GAAFET devices or the like) and be advantageous for subsequent formation of devices with short channel widths and long channel widths.

[0037] FIGS. **4A** and **4B**, following from FIGS. **3A** and **3B**, respectively, illustrate the semiconductor structure **100** in the short channel region **100A** and the long channel region **100B**, respectively, after anisotropically etching the sidewall spacer layer **168**, the alternating layer stack of nanosheets **130** and sacrificial layer **120**, and the substrate **110** to form a plurality of vertical recesses **155**. An anisotropic etch removes the lateral portions of the sidewall spacer layer **168** so that vertical portions of the sidewall spacer layer **168** remain as sidewall spacers **170** on the sidewalls of the dummy gate **150** and the hard mask **160**.

[0038] Next, a source/drain fin etch back process may be performed to anisotropically remove portions of the dielectric blocking layer **140** and the alternating layer stack of the nanosheets **130** and the sacrificial layer **120**, thereby forming fin features (primarily the channel regions of the transistors) under the dummy gate structures separated by the plurality of vertical recesses **155**. In various embodiments, these etch back processes may be performed as a single etch process or alternately as two or more etch processes. In certain embodiments, these etch back process may comprise one or more wet etch processes, plasma etch processes such as reactive ion etch (RIE) processes, or combinations of these or other etch processes. Sidewalls of the nanosheets **130** and the sacrificial layer **120** are exposed by the formation of the vertical recesses **155**. In various embodiments, nanosheets **130** of the layer stack in the short channel region **100A** have smaller widths than nanosheets **130** of the layer stack in the long channel region **100B**.

[0039] FIGS. **5**, **6**, **7A**, **7B**, and **8** follow from FIG. **3A** and illustrate cross-sectional views of intermediate stages of manufacturing for the semiconductor structure **100** performed in the short channel region **100A**. It should be understood that analogous intermediate stages of manufacturing are also performed on the semiconductor structure **100** in the long channel region **100B**.

[0040] In FIG. **5**, a lateral recess etch (cavity etch) is performed on the semiconductor structure **100** to selectively remove a portion of the sacrificial layers **120** relative to the nanosheets **130**. In other words, the sacrificial layers **120** are laterally recessed (or etched) to form a plurality of lateral recesses **165** (also referred to as indents) between layers of the nanosheets **130**. Although the recession of the sacrificial layers **120** may be described as lateral relative to the nanosheets **130**, it may be performed by a uniform (in other words, isotropic) recession or etching of the sacrificial layers **120**. In various embodiments, the lateral recess etch process comprises one or more isotropic etching process, such as one or more wet etch processes. In other embodiments, the lateral recess etch process includes plasma etch processes such as atomic layer etching processes as well as reactive ion etch (RIE) processes or combinations of these or other etch processes.

[0041] In FIG. **6**, following from FIG. **5**, an inner spacer layer **180** is deposited over the semiconductor structure **100**. In this step, the lateral recesses **165** (see above, FIG. **5**) are also filled with the material of the inner spacer layer **180**. The material of the inner spacer layer **180** may be made porous to reduce its dielectric constant in order to minimize the parasitic capacitance caused by the subsequently formed inner spacers (see below, FIGS. **7A-7B**) and thus the material of the inner spacer layer **180** may be a low-k dielectric material.

[0042] In some embodiments, the inner spacer layer **180** comprises silicon-containing dielectric

materials such as silicon nitride (SiN), silicon carbonitride (SiCN), silicon oxycarbonitride (SiOCN or SiOCNH), silicon boron carbonitride (SiBCN), the like, or a combination thereof. In one embodiment, the inner spacer layer **180** is a low-k dielectric material comprising SiOCNH with additional ionic bonds (e.g., carbon-hydrogen bonds) and fewer polar bonds than other silicon-containing dielectric materials, which may produce some porosity that may reduce its k value. The formation of the inner spacer layer **180** may be performed by deposition from a gas phase using, for example, chemical vapor deposition (CVD), plasma-enhanced CVD (PECVD), low-pressure CVD (LPCVD), physical vapor deposition (PVD), atomic layer deposition (ALD), other deposition processes, the like, or a combination thereof. For a plasma deposition process, a precursor gas mixture can be used including but not limited to silanes, hydrocarbons, fluorocarbons, or nitrogen containing compounds in combination with one or more dilution gases (e.g., argon, nitrogen, etc.) at a variety of pressure, power, flow and temperature conditions.

[0043] Next, in FIGS. 7A and 7B, an inner spacer etch back is performed on the semiconductor structure **100** to form inner spacers **190**. FIG. 7A illustrates a same cross-sectional view of the semiconductor structure **100** following from FIG. 6, and FIG. 7B illustrates another cross-sectional view of the semiconductor structure **100** perpendicular to the cross-sectional view of FIG. 7A. The inner spacer etch back removes a portion of the inner spacer layer **180** (see above, FIG. 6) to expose tips of the nanosheets **130**. The inner spacer etch back may remove the portions of the inner spacer layer **180** over the hard mask **160**, the sidewall spacers **170**, and the substrate **110**. Some portions of the inner spacer layer **180** are left between the layers of nanosheets **130** to form inner spacers **190**. These remaining portions may provide electrical insulation between the gate region and source/drain region that will be formed at later steps in fabrication.

[0044] In various embodiments, the inner spacer etch back process comprises one or more wet etch processes, plasma etch processes, reactive ion etch (RIE) processes, or combinations of these or other etch processes. In one or more embodiments, the sidewall spacer etch back process comprises an anisotropic etching process, for example, a RIE processes followed optionally by a short isotropic etching process to expose the layers of nanosheets **130**.

[0045] FIG. 7B illustrates another cross-sectional view of the semiconductor structure **100** along a cross section indicated by the dashed line **7b** in FIG. 7A. Conversely, FIG. 7A illustrates the cross-sectional view of the semiconductor structure **100** along a cross section indicated by the dashed line **7a** in FIG. 7B. In FIG. 7B, isolation regions such as shallow trench isolation (STI) regions **195** are visible, which may also be present in all of the previous embodiment illustrations (FIGS. 2, 3A, 3B, 4A, 4B, 5, and 6). In various embodiments, the STI regions **195** comprise an oxide such as silicon oxide (SiO) or silicon dioxide (SiO₂). The STI regions **195** serve to electrically isolate adjacent electronic components. With a completion of the inner spacer formation, the tips of the nanosheets **130**, the dielectric blocking layer **140**, and the inner spacers **190** are exposed and visible in the plurality of vertical recesses **155**. However, the sacrificial layers **120** are masked by the inner spacers **190**. Further, in FIG. 7B, the dummy gate **150** visible in FIG. 7A is masked by the sidewall spacer **170**. The sidewall spacer **170** is visible after a completion of removing portions of the inner spacer layer **180** (see above, FIG. 6).

[0046] FIG. 8, following from FIG. 7A and illustrating the same cross-sectional view, illustrates a cross-sectional view of the semiconductor structure **100** in the short channel region **100A** after source/drain region formation. In various embodiments, source/drain regions **210** are formed with a source/drain material. The source/drain material fills the plurality of vertical recesses **155** (see above, FIG. 7A) and covers the exposed tips of the nanosheets **130**. Any remaining unfilled portions of the lateral recesses **165** (see above, FIG. 5) may also be filled by the source/drain material. The formation of the source/drain regions **210** may be performed, for example, by epitaxial growth. In certain embodiments, the source/drain regions **210** formed with an epitaxial growth process comprise a faceted outer surface. In various embodiments, the semiconductor structure **100** is used to fabricate a pFET, and accordingly the source/drain material comprises a p-

type semiconductor, for example, silicon-germanium (SiGe). In various embodiments, the Ge concentration in the source/drain material is higher than that in the sacrificial layer **120**. In one or more embodiments, the SiGe for the source/drain material has a ratio of $\text{Si.sub.1-xGe.sub.x}$ where x is greater than 0.5. In other words, the Ge concentration of the SiGe in the source/drain regions **210** is 50% or greater, such as 50% to 70%. However, any suitable concentration of germanium may be used. The SiGe of the source/drain regions **210** may be further doped, such as with boron. In certain embodiments, the dopant concentration in the source/drain regions **210** is in a range of $1 \times 10^{20} \text{ cm}^{-3}$ to $5 \times 10^{20} \text{ cm}^{-3}$.

[0047] In certain embodiments, although not specifically illustrated, the semiconductor structure **100** further comprises one or more nFET structures adjacent to the pFET structures illustrated in FIGS. 2, 3A, 3B, 4A, 4B, 5, 6, 7A, and 7B. For the nFET structures, the semiconductor structure **100** may comprise a similar fin and dummy gate structure where a n-semiconductor (e.g., phosphorous-doped silicon) may be epitaxially grown from the tips of the nanosheets.

[0048] Next, FIGS. 9A and 9B illustrate cross-sectional views of the semiconductor structure **100** in the short channel region **100A** and the long channel region **100B**, respectively, after a dummy gate removal. FIG. 9A follows from FIG. 8 and FIG. 9B follows from FIG. 4B after the same process steps are performed as described above with respect to FIGS. 5-6, 7A, and 8. After the formation of the source/drain regions **210**, the remaining portion of the hard mask **160** and the dummy gates **150** are removed, such as with a dummy gate pull process. In certain embodiments, the dummy gate pull process is performed using a plasma etch process such as a reactive ion etch (RIE) process or the like.

[0049] In some embodiments, after the dummy gate pull and prior to a subsequent step of channel release (see below, FIG. 10A), an oxide removal process is performed to remove a surface oxide layer that may be present over the sacrificial layers **120**. For example, the oxide removal process may be a plasma-less process comprising exposing the semiconductor structure **100** to a process gas (such as a gas comprising, for example, hydrogen fluoride (HF)) in the absence of plasma. In an embodiment, the process gas comprises about 30% HF and 30% NH_3 in a carrier gas (e.g., argon (Ar)) and the process temperature is in a range of 35° C. to 80° C. The oxide removal process may be followed by a thermal treatment comprising heating the semiconductor structure **100** to a temperature in a range of 100° C. to 200° C. under an inert gas flow (such as an argon (Ar) flow).

[0050] FIGS. 10A, 10B, 11A, 11B, 11C, 11D, 12A, and 12B illustrate a method for channel release including selective oxidation of portions of source/drain regions **210** and a subsequent channel release selective to the oxidized portions of the source/drain regions. This may be advantageous by decreasing or preventing damage to the source/drain regions **210** without negatively affecting the channel release process performance, which can increase device yield and device performance. The selective oxidation and subsequent selective channel release may increase the margin of over-etching for channel release processes (such as processes including both short channels and long channels that may benefit from over-etching), which may reduce channel release process marginality and improve device yield. In various embodiments, the process steps described below with reference to 10A, 10B, 11A, 11B, 11C, 11D, 12A, and 12B are performed in a single process chamber (e.g., a vacuum chamber) and may be performed in situ. In some embodiments, the selective oxidation and channel release processes are plasma-free; in other words, the steps are performed without generation and/or use of plasma. Although embodiments of the selective oxidation and channel release are described using a semiconductor having short channel and long channel regions as an example, the selective oxidation and channel release methods may be used for the manufacturing suitable application (e.g., GAAFET, CFET, or the like) with or without short and/or long channel regions, and all such applications are within the scope of the disclosed embodiments.

[0051] Following from FIGS. 9A and 9B, FIGS. 10A and 10B illustrate cross-sectional views of the semiconductor structure **100** in the short channel region **100A** and the long channel region

100B, respectively, after a first channel release process. In various embodiments, the first channel release process removes most or all of the sacrificial material from the sacrificial layers **120** in the short channel region **100A**, and the first channel release process removes portions of the sacrificial material from the sacrificial layers **120** in the long channel region **100B**. This removal of the sacrificial layers **120** in the short channel region **100A** forms voids **220** between the nanosheets **130** and inner spacers **190**, and the removal of portions of the sacrificial layers **120** in the long channel region **100B** forms voids **230** between the nanosheets **130** and remaining portions of the sacrificial layers **120**. Although FIG. **9B** illustrates the voids **230** as a respective single void **230** being in the middle of each of the sacrificial layers **120**, any number of voids **230** may be formed with any sizes and positions between the nanosheets **130** and inner spacers **190**.

[0052] In some embodiments, the first channel release process is a plasma-less etch process that is a two-step process comprising a fluorocarbon pretreatment step and an etch step. Both steps may be advantageously performed in the absence of plasma. The fluorocarbon pretreatment comprises exposing the semiconductor structure **100** to a pretreatment gas comprising a fluorocarbon that may selectively passivate the material of the source/drain regions **210** (e.g., B-doped SiGe), and the etch step comprises exposing the semiconductor structure **100** to an etch gas comprising fluorine to etch the sacrificial layers **120** selectively, without causing damage to the source/drain regions **210**. In various embodiments, the fluorocarbon for the pretreatment gas comprises C.sub.2F.sub.6, C.sub.4F.sub.8, or a compound with a general chemical formula C.sub.xF.sub.y. In another embodiment, the fluorocarbon further comprises hydrogen (i.e., hydrofluorocarbon with a general chemical formula C.sub.xH.sub.yF.sub.z). However, any suitable protection method may be used to protect exposed surfaces of the source/drain regions **210** prior to removing the sacrificial layers **120** with an etch step.

[0053] In various embodiments, the etch step of the first channel release process is performed with an etch gas comprising fluorine. In one or more embodiments, the etch step is performed at a chamber pressure in a range of 10 mTorr to 500 mTorr, at a substrate temperature in a range of 0° C. to 55° C., and/or with a process gas comprising hydrogen fluoride (HF), fluorine (F.sub.2), chlorine trifluoride (ClF.sub.3), argon (Ar), nitrogen (N.sub.2), the like, or a combination thereof. In one embodiment, the process time is in a range of 10 seconds to 300 seconds. In one embodiment, the etch step is performed without exposing the semiconductor structure **100** to the fluorocarbon of the pretreatment gas or any other fluorocarbon. In another embodiment, the etch gas further comprises a fluorocarbon. In other embodiments, the fluorocarbon treatment is integrated into the etch step and the plasma-less etch process is performed as a single continuous process.

[0054] In some embodiments, after the first channel release process is complete, a post-etch thermal treatment is performed. This may be advantageous by removing etch byproducts and providing a clean process environment for subsequent processes, such as a selective oxidation and second channel release process (see below, FIGS. **11A-11B** and **12A-12B**). In various embodiments, the post-etch thermal treatment heats the semiconductor structure **100** to a temperature in a range of 100° C. to 200° C., at a chamber pressure in a range of 1000 mTorr to 5000 mTorr, and under a gas flow comprising argon (Ar), nitrogen (N.sub.2), the like, or a combination thereof.

[0055] Next, FIGS. **11A** and **11B** illustrate cross-sectional views of the semiconductor structure **100** in the short channel region **100A** and the long channel region **100B**, respectively, after a selective oxidation is performed on the source/drain regions **210**. The selective oxidation is used to oxidize portions (also referred to as surface layers) of the source/drain regions **210** adjacent to the inner spacers in the short channel region **100A**, thereby forming oxidized regions **240**. Due to their oxidation, the oxidized regions **240** may enable greater selectivity of a subsequent second channel release process (see below, FIGS. **12A-12B**) so that it removes remaining portions of the sacrificial layers **120** in the long channel region **100B** while reducing or preventing damage to the

source/drain regions **210** in the short channel region **100A**.

[0056] The selective oxidation may be performed at a chamber pressure in a range of 5 Torr to 90 mTorr, at a substrate temperature in a range of 25° C. to 300° C., and/or with a process gas comprising nitric oxide (NO), argon (Ar), nitrogen (N.sub.2), the like, or a combination thereof. In some embodiments, the selective oxidation is performed at a chamber pressure in a range of 5 Torr to 50 mTorr, at a substrate temperature in a range of 25° C. to 100° C., and with a process gas comprising 75% nitric oxide (NO) with argon (Ar) and nitrogen (N.sub.2). In one embodiment, the oxidation process time is in a range of 10 seconds to 300 seconds. Pressure and temperature may be tuned to produce a desired amount of oxidation for increasing selectivity of a subsequent etch process. The nitric oxide may enter the voids **220** and penetrate through the inner spacers **190** to oxidize portions of the source/drain regions **210** immediately adjacent to the inner spacers **190**, thereby forming oxidized regions **240**. Subsequently, these oxidized regions **240** may act as effective barriers to etchants during a second channel release process (see below, FIGS. **12A-12B**). The oxidation process is selective to the material of the source/drain regions **210** over the remaining material of the sacrificial layer **120**. This may be due to the greater ratio of germanium to silicon in the source/drain regions **210** (e.g., 55% or more Ge) compared with the sacrificial layer **120** (e.g., 25% or less Ge).

[0057] FIG. **11C** illustrates a graph of etched amount of silicon-germanium (SiGe) that has been oxidized before etching versus germanium percentage of the SiGe. Line **306** shows the relative change in etched amount of SiGe for an etch process after a selective oxidation process applied to SiGe with varying percentages of Si and Ge. Bar **302** represents SiGe with Ge percentage of 25% or less (e.g., SiGe in the sacrificial layer **120**) and bar **304** represents SiGe with Ge percentage of 55% or greater (e.g., SiGe in the source/drain regions **210**). As such, the selective oxidation process of line **306** may produce greater oxidation in the SiGe of the source/drain regions **210** compared with the SiGe of the sacrificial layer **120**. This may lead to enhancement of the selectivity of a subsequent etch to the less oxidized sacrificial layer **120** SiGe (bar **302**) compared with the more oxidized SiGe in the source/drain regions **210** (bar **304**, corresponding to the oxidized regions **240**; see above, FIG. **11A**).

[0058] FIG. **11D** illustrates experimental data of etched amount decrease versus germanium percentage of SiGe for a first process **310** and a second process **320**. The first process **310** may be an oxidation process (e.g., with 75% nitric oxide (NO), argon (Ar), and nitrogen (N.sub.2)) at lower pressure (e.g., 5 to 50 Torr) and lower temperature (e.g., 25° C. to 100° C.), and the second process **320** may be an oxidation process with the same gas mixture at higher pressure (e.g., 50 to 90 Torr) and higher temperature (e.g., 100° C. to 300° C.). As illustrated by FIG. **11D**, the first process **310** may be able to reduce the etched amount of SiGe with 55% Ge (e.g., SiGe in the source/drain regions **210**) by 16.5 nm compared to only reducing the etched amount of SiGe with 15% to 25% Ge (e.g., SiGe in the sacrificial layer **120**) by 5.1 nm to 3.5 nm. The second process **320** may be able to reduce the etched amount of SiGe with 55% Ge (e.g., SiGe in the source/drain regions **210**) by 1645 nm compared to only reducing the etched amount of SiGe with 15% (e.g., SiGe in the sacrificial layer **120**) by 4.0. As such, the first process **310** and the second process **320** may be advantageous for selective oxidation to produce oxidized regions **240** of the source/drain regions **210**, as described above with respect to FIG. **11A**.

[0059] Following from FIGS. **11A** and **11B**, FIGS. **12A** and **12B** illustrate cross-sectional views of the semiconductor structure **100** in the short channel region **100A** and the long channel region **100B**, respectively, after a second channel release process. In various embodiments, the second channel release process removes most or all of the remaining sacrificial material from the sacrificial layers **120** in the short channel region **100A**, expanding the voids **230** into voids **250** bounded in a cross-sectional view by the inner spacers **190** and the nanosheets **130**. The oxidized regions **240** protect the source/drain regions **210** from damage during the second channel release process due to the selectivity of the etch between the material of the sacrificial layers **120** and the

oxidized regions **240**, thereby increasing the margin of over-etching.

[0060] In some embodiments, the second channel release process is a plasma-less process performed with an etch gas comprising fluorine. In one or more embodiments, the etch step is performed at a chamber pressure in a range of 10 mTorr to 500 mTorr, at a substrate temperature in a range of 0° C. to 55° C., and/or with a process gas comprising hydrogen fluoride (HF), fluorine (F.sub.2), chlorine trifluoride (ClF.sub.3), argon (Ar), nitrogen (N.sub.2), the like, or a combination thereof. In one embodiment, the process time is in a range of 10 seconds to 300 seconds.

[0061] In some embodiments, after the second channel release process is complete, a(nother) post-etch thermal treatment is performed. This may be advantageous by removing etch byproducts and providing a clean process environment for subsequent processes, such as metal gate formation (see below, FIGS. **13A-13B**). In various embodiments, the post-etch thermal treatment heats the semiconductor structure **100** to a temperature in a range of 100° C. to 200° C., at a chamber pressure in a range of 1000 mTorr to 5000 mTorr, and under a gas flow comprising argon (Ar), nitrogen (N.sub.2), the like, or a combination thereof.

[0062] Next, FIGS. **13A** and **13B** illustrate cross-sectional views of the semiconductor structure **100** in the short channel region **100A** and the long channel region **100B**, respectively after a replacement gate formation, such as a high-k metal gate (HKMG). The channel release process (see above, FIGS. **10A-12B**) frees up space previously occupied by the sacrificial layers **120**. This created void space (e.g., voids **220** and voids **250**) may be filled with gate dielectric (e.g., high-k dielectric materials) and gate electrode through the HKMG formation in both the short channel region **100A** and the long channel region **100B**. As an example of a HKMG formation process, first, a high-k dielectric layer **260** is deposited. In some embodiments, the high-k dielectric layer **260** comprises hafnium dioxide (HfO.sub.2), Hf.sub.xSi.sub.yO.sub.zN.sub.w, or the like. The high-k dielectric layer **260** may be deposited using appropriate deposition techniques such as vapor deposition including atomic layer deposition (ALD), chemical vapor deposition (CVD), physical vapor deposition (PVD), as well as other plasma processes such as plasma enhanced CVD (PECVD), sputtering, the like, or a combination thereof. In various embodiments, an optional insulating layer, such as a silicon oxide layer, is formed beneath the high-k dielectric layer **260**.

[0063] Next, over the high-k dielectric layer **260**, a replacement metal gate (RMG) material **270** is deposited to fill the remainder of the void space and complete the HKMG formation. In various embodiments, the RMG material **270** comprises a combination of several layers, including a workfunction metal and a metallic fill material. In some embodiments, the workfunction metal of the RMG material comprises titanium nitride (TiN), tantalum nitride (TaN), metal alloys such as AlC, TiAl and TiAlC, the like, or a combination thereof. Metal deposition is continued till the remaining recesses are filled with excess metallic fill material. In some embodiments, the metallic fill material comprise a low resistivity metal such as tungsten (W), copper (Cu), cobalt (Co), aluminum (Al), the like, or a combination thereof. In some embodiments, the RMG material **270** may be deposited using a highly conformal process such as an atomic layer deposition (ALD) process. However, any suitable process may be used to form the RMG material **270**. After the deposition steps, any excess metal may be removed by a planarizing process such as chemical mechanical planarization (CMP). Subsequently, middle-of-line (MOL)/back-end-of-line (BEOL) processes may be performed, such as to form various inter-metal dielectric layers, gate and source/drain conductive contacts, interconnect structures, the like, or a combination thereof, to the GAAFET device(s) of the semiconductor structure **100**. Similar processes may be used for other FET devices such as CFETs and are within the scope of the disclosed embodiments.

[0064] FIG. **14** illustrates a flow chart diagram of a method **1000** for processing a substrate, in accordance with some embodiments. In step **1002**, a recess is formed by etching a sacrificial layer, as described above with respect to FIG. **5**. The sacrificial layer is between a lower nanosheet and an upper nanosheet. In step **1004**, an inner spacer is formed in the recess, as described above with respect to FIGS. **6** and **7A**.

[0065] In step **1006**, a source/drain region is formed adjacent the lower nanosheet, the inner spacer, and the upper nanosheet, as described above with respect to FIG. **8**. In step **1008**, the sacrificial layer is removed with an etch process, as described above with respect to FIG. **10A**. In step **1010**, an oxidation is performed on a portion of the source/drain region through the inner spacer, as described above with respect to FIG. **11A**.

[0066] FIG. **15** illustrates a flow chart diagram of a method **1100** for processing a substrate, in accordance with some embodiments. In step **1102**, a first layer stack and a second layer stack are formed, as described above with respect to FIGS. **2-4B**. The first layer stack comprises alternating layers of first nanosheets and first sacrificial layers, the second layer stack comprises alternating layers of second nanosheets and second sacrificial layers, and the second layer stack is wider than the first layer stack.

[0067] In step **1104**, a first channel release process is performed on the first layer stack and the second layer stack, as described above with respect to FIGS. **10A-10B**. The first channel release process removes the first sacrificial layers from the first layer stack and a portion of the second sacrificial layers from the second layer stack.

[0068] In step **1106**, an oxidation is performed on a portion of a source/drain region, as described above with respect to FIG. **11A**. The source/drain region is adjacent the first layer stack. In step **1108**, a second channel release process is performed on the second layer stack, as described above with respect to FIGS. **12A-12B**. The second channel release process removes the remaining portion of the second sacrificial layers.

[0069] FIG. **16** illustrates a flow chart diagram of a method **1200** for processing a substrate, in accordance with some embodiments. In step **1202**, a first layer stack and a second layer stack are formed, as described above with respect to FIGS. **2-4B**. The first layer stack and the second layer stack each comprise alternating layers of silicon (Si) layers and silicon-germanium (SiGe) layers, and the first layer stack is wider than the second layer stack.

[0070] In step **1204**, indents are formed between the Si layers of the first layer stack and the second layer stack by etching a portion of the SiGe layers selectively to the Si layers, as described above with respect to FIG. **5**. In step **1206**, respective inner spacers are formed in the indents of the first layer stack and the second layer stack, as described above with respect to FIGS. **6-7A**. In step **1208**, a first source/drain region is epitaxially grown adjacent the first layer stack and a second source/drain region is epitaxially grown adjacent the second layer stack, as described above with respect to FIG. **8**.

[0071] In step **1210**, a portion of the SiGe layers of the first layer stack and the SiGe layers of the second layer stack are removed with a first etch process, as described above with respect to FIGS. **10A-10B**. In step **1212**, oxidized regions are formed in the second source/drain region through respective inner spacers of the second layer stack, as described above with respect to FIG. **11A**. In step **1214**, the remaining portion of the SiGe layers of the first layer stack are removed with a second etch process, as described above with respect to FIGS. **12A-12B**.

[0072] Example embodiments of the disclosure are summarized here. Other embodiments can also be understood from the entirety of the specification as well as the claims filed herein.

[0073] Example 1. A method of processing a substrate, the method including: forming a recess by etching a sacrificial layer, the sacrificial layer being between a lower nanosheet and an upper nanosheet; forming an inner spacer in the recess; forming a source/drain region adjacent the lower nanosheet, the inner spacer, and the upper nanosheet; removing the sacrificial layer with an etch process; and performing an oxidation on a portion of the source/drain region through the inner spacer.

[0074] Example 2. The method of example 1, where the sacrificial layer and the source/drain region include silicon-germanium (SiGe).

[0075] Example 3. The method of example 2, where the germanium percentage of the sacrificial layer is 25% or less.

[0076] Example 4. The method of one of examples 2 or 3, where the germanium percentage of the source/drain region is 55% or greater.

[0077] Example 5. The method of one of examples 1 to 4, where performing the oxidation includes using a gas including nitric oxide (NO), argon (Ar), and nitrogen (N.sub.2).

[0078] Example 6. The method of one of examples 1 to 5, where the oxidation is performed at a pressure in a range of 5 Torr to 90 Torr.

[0079] Example 7. The method of one of examples 1 to 6, where the oxidation is performed at a temperature in a range of 25° C. to 300° C.

[0080] Example 8. A method of processing a substrate, the method including: forming a first layer stack and a second layer stack, the first layer stack including alternating layers of first nanosheets and first sacrificial layers, the second layer stack including alternating layers of second nanosheets and second sacrificial layers, the second layer stack being wider than the first layer stack; performing a first channel release process on the first layer stack and the second layer stack, the first channel release process removing the first sacrificial layers from the first layer stack and a portion of the second sacrificial layers from the second layer stack; performing an oxidation on a portion of a source/drain region, the source/drain region being adjacent the first layer stack; and performing a second channel release process on the second layer stack, the second channel release process removing the remaining portion of the second sacrificial layers.

[0081] Example 9. The method of example 8, where the oxidation includes a gas with 75% nitric oxide (NO).

[0082] Example 10. The method of example 9, where the gas further includes argon (Ar) and nitrogen (N.sub.2).

[0083] Example 11. The method of one of examples 8 to 10, where the oxidation is performed at a pressure in a range of 5 Torr to 50 Torr.

[0084] Example 12. The method of one of examples 8 to 11, where the oxidation is performed at a temperature in a range of 25° C. to 100° C.

[0085] Example 13. The method of one of examples 8 to 12, where the source/drain region includes a higher germanium percentage than the second sacrificial layers.

[0086] Example 14. The method of example 13, where the source/drain region includes 55% or more of germanium.

[0087] Example 15. The method of one of examples 13 or 14, where the second sacrificial layers include 25% or less of germanium.

[0088] Example 16. The method of one of examples 8 to 15, further including forming respective inner spacers adjacent the first sacrificial layers in the first layer stack, and where the portion of the source/drain region is oxidized through the inner spacers.

[0089] Example 17. A method of processing a substrate, the method including: forming a first layer stack and a second layer stack, the first layer stack and the second layer stack each including alternating layers of silicon (Si) layers and silicon-germanium (SiGe) layers, the first layer stack being wider than the second layer stack; forming indents between the Si layers of the first layer stack and the second layer stack by etching a portion of the SiGe layers selectively to the Si layers; forming respective inner spacers in the indents of the first layer stack and the second layer stack; epitaxially growing a first source/drain region adjacent the first layer stack and a second source/drain region adjacent the second layer stack; removing a portion of the SiGe layers of the first layer stack and the SiGe layers of the second layer stack with a first etch process; forming oxidized regions in the second source/drain region through respective inner spacers of the second layer stack; and removing the remaining portion of the SiGe layers of the first layer stack with a second etch process.

[0090] Example 18. The method of example 17, where forming the oxidized regions includes flowing a gas including 75% nitric oxide (NO) at a pressure in a range of 5 Torr to 50 Torr and a temperature in a range of 25° C. to 100° C.

[0091] Example 19. The method of one of examples 17 or 18, where the first etch process, the forming the oxidized regions, and the second etch process are performed in situ.

[0092] Example 20. The method of one of examples 17 to 19, where the first etch process, the forming the oxidized regions, and the second etch process are performed without plasma.

[0093] While this invention has been described with reference to illustrative embodiments, this description is not intended to be construed in a limiting sense. Various modifications and combinations of the illustrative embodiments, as well as other embodiments of the invention, will be apparent to persons skilled in the art upon reference to the description. It is therefore intended that the appended claims encompass any such modifications or embodiments.

Claims

1. A method of processing a substrate, the method comprising: forming a recess by etching a sacrificial layer, the sacrificial layer being between a lower nanosheet and an upper nanosheet; forming an inner spacer in the recess; forming a source/drain region adjacent the lower nanosheet, the inner spacer, and the upper nanosheet; removing the sacrificial layer with an etch process; and performing an oxidation on a portion of the source/drain region through the inner spacer.
2. The method of claim 1, wherein the sacrificial layer and the source/drain region comprise silicon-germanium (SiGe).
3. The method of claim 2, wherein the germanium percentage of the sacrificial layer is 25% or less.
4. The method of claim 2, wherein the germanium percentage of the source/drain region is 55% or greater.
5. The method of claim 1, wherein performing the oxidation comprises using a gas comprising nitric oxide (NO), argon (Ar), and nitrogen (N.sub.2).
6. The method of claim 1, wherein the oxidation is performed at a pressure in a range of 5 Torr to 90 Torr.
7. The method of claim 1, wherein the oxidation is performed at a temperature in a range of 25° C. to 300° C.
8. A method of processing a substrate, the method comprising: forming a first layer stack and a second layer stack, the first layer stack comprising alternating layers of first nanosheets and first sacrificial layers, the second layer stack comprising alternating layers of second nanosheets and second sacrificial layers, the second layer stack being wider than the first layer stack; performing a first channel release process on the first layer stack and the second layer stack, the first channel release process removing the first sacrificial layers from the first layer stack and a portion of the second sacrificial layers from the second layer stack; performing an oxidation on a portion of a source/drain region, the source/drain region being adjacent the first layer stack; and performing a second channel release process on the second layer stack, the second channel release process removing the remaining portion of the second sacrificial layers.
9. The method of claim 8, wherein the oxidation comprises a gas with 75% nitric oxide (NO).
10. The method of claim 9, wherein the gas further comprises argon (Ar) and nitrogen (N.sub.2).
11. The method of claim 8, wherein the oxidation is performed at a pressure in a range of 5 Torr to 50 Torr.
12. The method of claim 8, wherein the oxidation is performed at a temperature in a range of 25° C. to 100° C.
13. The method of claim 8, wherein the source/drain region comprises a higher germanium percentage than the second sacrificial layers.
14. The method of claim 13, wherein the source/drain region comprises 55% or more of germanium.
15. The method of claim 13, wherein the second sacrificial layers comprise 25% or less of germanium.

16. The method of claim 8, further comprising forming respective inner spacers adjacent the first sacrificial layers in the first layer stack, and wherein the portion of the source/drain region is oxidized through the inner spacers.

17. A method of processing a substrate, the method comprising: forming a first layer stack and a second layer stack, the first layer stack and the second layer stack each comprising alternating layers of silicon (Si) layers and silicon-germanium (SiGe) layers, the first layer stack being wider than the second layer stack; forming indents between the Si layers of the first layer stack and the second layer stack by etching a portion of the SiGe layers selectively to the Si layers; forming respective inner spacers in the indents of the first layer stack and the second layer stack; epitaxially growing a first source/drain region adjacent the first layer stack and a second source/drain region adjacent the second layer stack; removing a portion of the SiGe layers of the first layer stack and the SiGe layers of the second layer stack with a first etch process; forming oxidized regions in the second source/drain region through respective inner spacers of the second layer stack; and removing the remaining portion of the SiGe layers of the first layer stack with a second etch process.

18. The method of claim 17, wherein forming the oxidized regions comprises flowing a gas comprising 75% nitric oxide (NO) at a pressure in a range of 5 Torr to 50 Torr and a temperature in a range of 25° C. to 100° C.

19. The method of claim 17, wherein the first etch process, the forming the oxidized regions, and the second etch process are performed in situ.

20. The method of claim 17, wherein the first etch process, the forming the oxidized regions, and the second etch process are performed without plasma.
